DS	**Session**	Searching	**Question Information**	Level 1 Challenge 8

Problem Description:
Ram's first job is a out of university is for a student's book publisher. They are currently publishing a line of "intelligent libs" books, and they need you to check their print-copy. Write a program that will take a sentence with placeholders in it, and substitute in for those placeholders words from the lists of example words the publisher has provided.

Your goal will be to substitute the appropriate words into the sentence from the list of provided words TWICE (keeping track of the words you have used) where:

[N] = nouns
[AV] = adverbs
[V] = verbs
[AJ] = adjectives

Input Format:
Line one will always be the line with the story. The order of the Nouns, Adverbs, Verbs and Adjectives will always be the same. Input stops when it reaches the word END alone on a line. You are guaranteed to always have enough words to substitute into the sentence TWICE (not repeating any words). The number of words given under each section will be random.

Output Format:
Print the output in a separate lines contains, Always take the TOP-most UNUSED word from a group when substituting. Output the final sentences when all substitutions are done.

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
There once was a [AJ] [N] from [N] who [AV] [V] all day.
NOUNS
squirrel
London
tree
lettuce
ADVERBS
silently
quickly
```

Test Case 2

INPUT (STDIN)

```
There once was a [AJ] [N] from [N] who [AV] [V] all day.
NOUNS
singer
chennai
hotel
chettinad
ADVERBS
silently
quickly
```

```
happily
VERBS
skipped
ran
climbs
ADJECTIVES
delightful
homely
quain
```

EXPECTED OUTPUT

```
There once was a delightful squirrel from London who
silently skipped all day.
There once was a homely tree from lettuce who quickly ran
all day.
```

```
happily
sadly
VERBS
skipped
reads
ran
sings
climbs
ADJECTIVES
delightful
homely
surprisingly
```

EXPECTED OUTPUT

```
There once was a delightful singer from chennai who silently
skipped all day.
There once was a homely hotel from chettinad who quickly
reads all day.
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
int cl[CMDS]
```

Test Case 2

KEYWORD

```
char *lists[CMDS][MAXWORDS];
```

Test Case 3

KEYWORD

```
char *tokens[TOKENS]={"[N]","[AV]","
[V]","[AJ]"};
```

Test Case 4

KEYWORD

```
char *cmds[CMDS]=
{"NOUNS","ADVERBS","VERBS","ADJECTIVE
S","END"};
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

```
3
```

Test Case 2

TOKEN COUNT

```
488
```

Test Case 3

NLOC

```
50
```

Code

solution.c

```c
#include <stdio.h>
#include <string.h>

#define CMDS 4
#define MAXWORDS 50
#define BUFLEN 200

char *cmds[CMDS] = {"NOUNS", "ADVERBS", "VERBS", "ADJECTIVES"};
char *tokens[4] = {"[N]", "[AV]", "[V]", "[AJ]"};
char lists[CMDS][MAXWORDS][50];
int counts[CMDS] = {0};
int ptrs[CMDS] = {0};

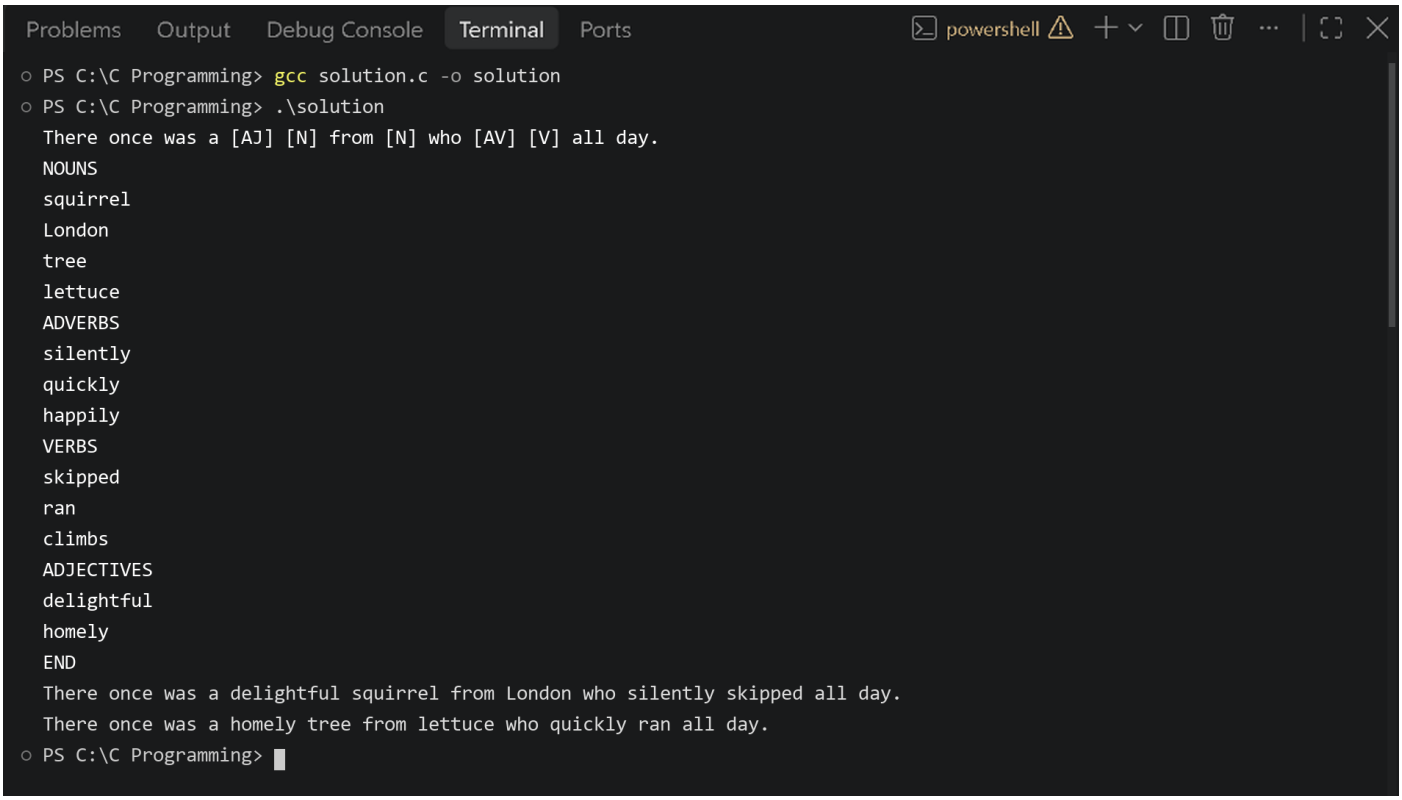
void replaceToken(char *src, const char *token, const char *rep, char *dest) {
    char *pos = strstr(src, token);
    if (!pos) { strcpy(dest, src); return; }
    int len = pos - src;
    strncpy(dest, src, len);
    dest[len] = ' ';
    strcat(dest, rep);
    strcat(dest, pos + strlen(token));
}

int main() {
    char story[500];
    if (!fgets(story, sizeof(story), stdin)) return 0;
    story[strcspn(story, "\r\n")] = '\0';

    char line[BUFLEN];
    int curCmd = -1;
    while (fgets(line, sizeof(line), stdin)) {
        line[strcspn(line, "\r\n")] = '\0';
        if (strcmp(line, "END") == 0) break;
        int foundCmd = -1;
        for (int i = 0; i < CMDS; i++) {
            if (strcmp(line, cmds[i]) == 0) { foundCmd = i; break; }
        }
        if (foundCmd != -1) curCmd = foundCmd;
        else if (curCmd != -1) {
            strcpy(lists[curCmd][counts[curCmd]++], line);
        }
    }

    for (int run = 0; run < 2; run++) {
        char result[1000];
        strcpy(result, story);
        for (int i = 0; i < 4; i++) {
            char temp[1000];
            while (strstr(result, tokens[i])) {
                replaceToken(result, tokens[i], lists[i][ptrs[i]++], temp);
                strcpy(result, temp);
            }
        }
        printf("%s\n", result);
    }
    return 0;
}
```

Output


Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\C Programming> gcc solution.c -o solution
PS C:\C Programming> .\solution
There once was a [AJ] [N] from [N] who [AV] [V] all day.
NOUNS
squirrel
London
tree
lettuce
ADVERBS
silently
quickly
happily
VERBS
skipped
ran
climbs
ADJECTIVES
delightful
homely
END
There once was a delightful squirrel from London who silently skipped all day.
There once was a homely tree from lettuce who quickly ran all day.
PS C:\C Programming>